

Question number	Answer	Additional guidance	Mark
2(a)	$1.431 \times 10^{15} / 1.43 \times 10^{15} / 1.4 \times 10^{15}$ (1)	ACCEPT $1.431 \cdot 10^{15} / 1.43 \cdot 10^{15} / 1.4 \cdot 10^{15}$	(1)

Question number	Answer	Mark
2(b)	The correct answer is D <i>A is incorrect because haemoglobin is not fibrous, it is globular</i> <i>B is incorrect because haemoglobin has quaternary structure</i> <i>C is incorrect because haemoglobin is not fibrous</i>	(1)

Question number	Answer	Additional guidance	Mark
2(c)(i)	A description that includes the following points: - embryonic development: α, β and γ subunits increase (1) - fetal development: constant levels of α, (constant levels and then) increase presence β, (short increase and then decreases) decreasing presence of γ (1) - baby: presence of α remains the same, increased presence of β (which levels off), decrease in γ (and then levels off) (1) OR α subunit increases during embryonic development and then remain constant (1) β subunit increases throughout embryonic and fetal development and in the baby (and then levels off) (1) γ subunit increases during embryonic development and then falls during fetal development and in the baby (and then levels off) (1)	DO NOT piece together or mix mps from both sets ACCEPT answers in weeks not stages throughout provided correct values are quoted and fall within: embryonic : up to 8 fetal : between 8 and 39 / 40 baby : between 0 and 42 ACCEPT increases and then at birth increases and levels off	(3)

Question number	Answer	Additional guidance	Mark
2(c)(ii)	An explanation that includes the following points: • fetal haemoglobin has higher affinity (than adult) (1) • so that oxygen will leave the adult haemoglobin and bind to fetal haemoglobin (1) • at the same partial pressure of oxygen (1)	Penalise baby for fetus once ACCEPT converse ACCEPT stronger {binding / association} (with oxygen) Y globin has higher affinity than β globin IGNORE easier / better / faster ACCEPT so oxygen will pass from mother's blood into fetus' blood so that the fetal haemoglobin will be more saturated (with oxygen than the adult) ACCEPT same oxygen concentration	(2)